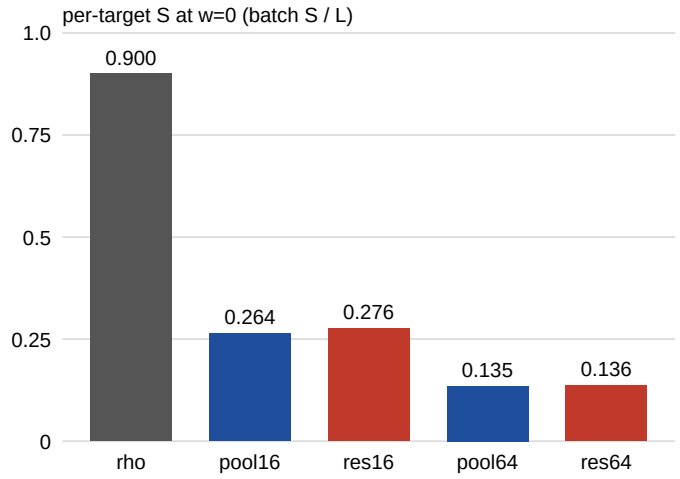
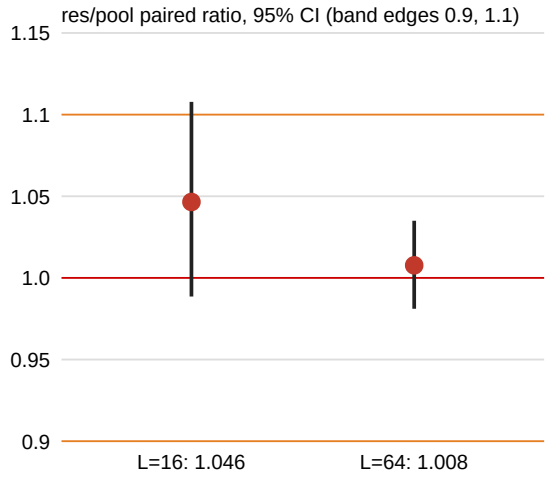
Index-calculus structure in the batch vs multi-target rho (ecbench, p36)

Protocol `PROTOCOL.md` frozen before data (Amendments 1-2: two code fixes before any valid data; the void attempts are kept in `aborted-attempt1/`, `aborted-attempt2/`). 320 / 320 records verified; `ecbench verify --replay 12`: audit OK, 12 replays reproduced, 0 problems. Local L0 (counts deterministic); an L2 replay is owed before any benchmarks row.

`ic.residual_batch` is `rho.multitarget_pooled` (same companions, walker, table and charging) plus one public base point P0 of unknown log in every relation, the residual collector's B = 1 column; log P0 was solved on the first target of every record and checked.

Claim P (parity, frozen rule: 95% interval contains 1 and lies in [0.9, 1.1] at w = 0, 0.1, 1)





batch L	res / pool (all three weights)	95% interval	rule
16	1.0465	[0.9886, 1.1078]	**fails** (upper edge above 1.1)
64	1.0077	[0.9811, 1.0350]	**holds**

The extra unknown costs about one extra collision per batch: 4.7% at L = 16, 0.8% at L = 64. At L = 64 the index-calculus structure is at parity with multi-target rho; at L = 16 its interval reaches past the frozen 1.1 edge, so parity is not established there.

Claim D (per-target cost against single-target rho)

arm	L	S w=0	per target	S w=0.1	per target	S w=1	per target	single rho / per target (w=0)
rho	1	0.900	0.900	0.990	0.990	1.799	1.799	1.00x
pool16	16	4.220	0.264	4.641	0.290	8.430	0.527	3.41x
res16	16	4.416	0.276	4.857	0.304	8.822	0.551	3.26x
pool64	64	8.654	0.135	9.516	0.149	17.277	0.270	6.66x
res64	64	8.720	0.136	9.589	0.150	17.410	0.272	6.61x

Per target, both batch methods are 3.3-3.4x (L = 16) and 6.6-6.7x (L = 64) cheaper than single-target rho at every weight. This is the multi-target regime: the matched reference for a batch is multi-target rho, against which the index-calculus structure is at parity (L = 64), not ahead. Single-target index calculus remains 1.6x rho (earlier studies).

Scripts: `scripts/analyse.sh`; derived files in `analysis/`; raw records in the artifact store.